

Reviewer Pack — Koszul Complex Lower Bound for SAT Refutation Length

Entry 9e0efb5891ad · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Koszul Complex Lower Bound for SAT Refutation Length

Entry ID: 9e0efb5891ad **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-30 19:46:38 UTC

1. Conjecture

Field A (mathematical branch): Homological Algebra (specifically, Koszul complexes) **Field B** (complexity object): SAT (Satisfiability)

Statement:

For every n -variables SAT instance φ with m clauses, the minimal number of generators in the Koszul complex associated with φ is $O(m^{(2/3)n}(1/6))$.

Rationale (proposer's reasoning):

Koszul complexes provide a homological approach to studying algebraic structures related to polynomial equations. Their combinatorial properties may reveal hidden complexity in satisfiability problems, offering insights into proof length and refutation complexity.

Taxonomy category: Homological_algebra_SAT (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 3bd7764b4050c3b3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 90% of the generated n -variables SAT instances with m clauses, the ratio of the computed minimal number of generators to $m^{(2/3)n}(1/6)$ is less than or equal to 1.05.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.90	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	0.95	UNCERTAIN	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
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4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'Koszul complex' AND 'SAT problem' AND 'refutation length' - 'Homological algebra' AND 'satisfiability problem' AND 'Koszul complex lower bound' - $0(m^{(2/3)})$ AND $0(n^{(1/6)})$ AND 'Koszul complex'

Top relevant hits considered: - [http://arxiv.org/abs/math/0012264v2] Koszul duality and equivalences of categories - [http://arxiv.org/abs/2604.20177v2] Koszul Duality for Quadratic Monomial Algebras - [http://arxiv.org/abs/math/0301233v1] Noncommutative Koszul filtrations - [http://arxiv.org/abs/math/0511307v2] A note on Rees algebras and the MFMC property - [http://arxiv.org/abs/math/0307281v1] Ancestor ideals of vector spaces of forms, and level algebras - [http://arxiv.org/abs/0710.2748v2] Algebraic curves for commuting elements in the q -deformed Heisenberg algebra - [http://arxiv.org/abs/2308.03822v1] Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo - [http://arxiv.org/abs/2509.08054v1] GW250114: testing Hawking's area law and the Kerr nature of black holes

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_sat_instance(n, m):
        clauses = []
        for _ in range(m):
            clause = [random.randint(1, n), random.randint(-n, -1)]
            clauses.append(clause)
        return clauses

    def koszul_complex_size(clauses):
        # Simplified version of Koszul complex size calculation
        return len(clauses) ** 2 / 3 * (n ** 1 / 6)

    n = random.randint(5, 40)
    m = int(n * 0.2) # Fixed clause density for simplicity

    clauses = generate_sat_instance(n, m)
    generators = koszul_complex_size(clauses)

    metric_value = generators
    instances_tested = 1
    n_max = n
    conjecture_holds = True if generators <= (m ** (2/3) * n ** (1/6)) * 1.05 else False
    counterexample = "" if conjecture_holds else "Koszul complex size exceeds bound"
```

```

return {
    "metric_name": "Koszul Complex Generators",
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.9:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"Koszul complex size exceeds bound\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'metric_name': 'Koszul Complex Generators', 'metric_value': 0.3888888888888889, 'instances_tested': 1
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 34.72222222222223, 'instances_tested': 1
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 20.444444444444443, 'instances_tested': 1
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 0.5, 'instances_tested': 1, 'n_max': 10
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 103.44444444444443, 'instances_tested': 1
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 8.0, 'instances_tested': 1, 'n_max': 10
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 98.0, 'instances_tested': 1, 'n_max': 10
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 9.5, 'instances_tested': 1, 'n_max': 10
TRIAL: {'metric_name': 'Koszul Complex Generators', 'metric_value': 8.5, 'instances_tested': 1, 'n_max': 10

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96c0ace8.py", line 71, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~^~~~~~

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The conjecture was falsified as multiple trials showed that the Koszul complex size exceeds the bound. | next: Investigate further to understand why the Koszul complex size is exceeding the bound and explore alternative bounds or methods for SAT refutation length.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15775
2	preregistration	ollama_remote	glm4:latest	0	0	11136
3	novelty	ollama_remote	glm4:latest	0	0	18593
4	novelty	ollama_remote	glm4:latest	0	0	19330
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14063
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20028
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7631
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13499
9	judge	ollama_remote	glm4:latest	0	0	9415

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 129471 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/9e0efb5891ad.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/9e0efb5891ad.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/9e0efb5891ad.tar.gz` (if generated)